UNIVERSITI MALAYA

NFMN

SIM 1005 / SIN 1002

Fundamentals of Spreadsheets / Introduction to Worksheet

Lecture 2: Excel Built-in Functions

By

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Objectives

- To classify types of Excel built-in functions
- To learn specific roles of the built-in functions for further utilization

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Types of Built-in Functions

- Basic Numeric Functions
- Basic Mathematical Functions
- Conditional Sums
- Advanced Mathematical Functions
- Exponents & Logarithms
- Trigonometry Functions
- Rounding Functions
- Random Numbers
- Matrix Functions

Original notes by NFM Noor

Basic Numeric Operations

ABS	Returns the absolute value (ie. the modulus) of a supplied number
SIGN	Returns the sign (+1, -1 or 0) of a supplied number
GCD	Returns the Greatest Common Divisor of two or more supplied numbers
LCM	Returns the Least Common Multiple of two or more supplied numbers

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Basic Mathematical Functions

SUM	Returns the sum of a supplied list of numbers
PRODUCT	Returns the product of a supplied list of numbers
POWER	Returns the result of a given number raised to a supplied power
SQRT	Returns the positive square root of a given number
QUOTIENT	Returns the integer portion of a division between two supplied numbers

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MOD	Returns the remainder from a division between two supplied numbers
AGGREGATE	Performs a specified calculation (eg. the sum, product, average, etc.) for a list or database, with the option to ignore hidden rows and error values *(New in Excel 2010)*
SUBTOTAL	Performs a specified calculation (eg. the sum, product, average, etc.) for a supplied set of values

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Conditional Sums

SUMIF	Adds the cells in a supplied range, that satisfy a given criteria
SUMIFS	Adds the cells in a supplied range, that satisfy multiple criteria *(New in Excel 2007)*

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Advanced Mathematical Functions

SUMPRODUCT	Returns the sum of the products of corresponding values in two or more supplied arrays
SUMSQ	Returns the sum of the squares of a supplied list of numbers
SUMX2MY2	Returns the sum of the difference of squares of corresponding values in two supplied arrays
SUMX2PY2	Returns the sum of the sum of squares of corresponding values in two supplied arrays
SUMXMY2	Returns the sum of squares of differences of corresponding values in two supplied arrays
SERIESSUM	Returns the sum of a power series

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Exponents & Logarithms

EXP	Returns *e* raised to a given power
LN	Returns the natural logarithm of a given number
LOG	Returns the logarithm of a given number, to a specified base
LOG10	Returns the base 10 logarithm of a given number

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Trigonometry Functions

PI	Returns the constant value of pi
SQRTPI	Returns the square root of a supplied number multiplied by pi
DEGREES	Converts Radians to Degrees
RADIANS	Converts Degrees to Radians
COS	Returns the Cosine of a given angle
ACOS	Returns the Arccosine of a number
COSH	Returns the hyperbolic cosine of a number
ACOSH	Returns the inverse hyperbolic cosine of a number

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SEC	Returns the secant of an angle *(New in Excel 2013)*
SECH	Returns the hyperbolic secant of an angle *(New in Excel 2013)*
SIN	Returns the Sine of a given angle
ASIN	Returns the Arcsine of a number
SINH	Returns the Hyperbolic Sine of a number
ASINH	Returns the Inverse Hyperbolic Sine of a number
CSC	Returns the cosecant of an angle *(New in Excel 2013)*
CSCH	Returns the hyperbolic cosecant of an angle *(New in Excel 2013)*

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ATAN	Returns the Arctangent of a given number
ATAN2	Returns the Arctangent of a given pair of x and y coordinates
TANH	Returns the Hyperbolic Tangent of a given number
ATANH	Returns the Inverse Hyperbolic Tangent of a given number
COT	Returns the cotangent of an angle *(New in Excel 2013)*
COTH	Returns the hyperbolic cotangent of an angle *(New in Excel 2013)*
ACOT	Returns the arccotangent of a number *(New in Excel 2013)*
ACOTH	Returns the hyperbolic arccotangent of a number *(New in Excel 2013)*

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Rounding Functions

CEILING	Rounds a number away from zero (ie. rounds a positive number up and a negative number down), to a multiple of significance
CEILING.PRECISE	Rounds a number up, regardless of the sign of the number, to a multiple of significance *(New in Excel 2010)*
ISO.CEILING	Rounds a number up, regardless of the sign of the number, to a multiple of significance. *(New in Excel 2010)*
CEILING.MATH	Rounds a number up to the nearest integer or to the nearest multiple of significance *(New in Excel 2013)*
EVEN	Rounds a number away from zero (ie. rounds a positive number up and a negative number down), to the next even number

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FLOOR	Rounds a number towards zero, (ie. rounds a positive number down and a negative number up), to a multiple of significance
FLOOR.PRECISE	Rounds a number down, regardless of the sign of the number, to a multiple of significance *(New in Excel 2010)*
FLOOR.MATH	Rounds a number down, to the nearest integer or to the nearest multiple of significance *(New in Excel 2013)*
INT	Rounds a number down to the next integer
MROUND	Rounds a number up or down, to the nearest multiple of significance

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ODD	Rounds a number away from zero (ie. rounds a positive number up and a negative number down), to the next odd number
ROUND	Rounds a number up or down, to a given number of digits
ROUNDDOWN	Rounds a number towards zero, (ie. rounds a positive number down and a negative number up), to a given number of digits
ROUNDUP	Rounds a number away from zero (ie. rounds a positive number up and a negative number down), to a given number of digits
TRUNC	Truncates a number towards zero (ie. rounds a positive number down and a negative number up), to the next integer.

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Random Numbers

RAND	Returns a random number between 0 and 1
RANDBETWEEN	Returns a random number between two given integers

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Matrix Functions

MDETERM	Returns the matrix determinant of a supplied array
MINVERSE	Returns the matrix inverse of a supplied array
MMULT	Returns the matrix product of two supplied arrays
MUNIT	Returns the unit matrix for a specified dimension *(New in Excel 2013)*

Original notes by NFM Noor

Summary

- Excel built-in functions are tools for data and numerical operations.
- Conditional sums are functions for conditional arguments.
- Advanced mathematical functions are for series and array data operations.
- Rounding functions are for data precision and accuracy.
- Matrix functions are for basic matrix operations.

Original notes by NFM Noor